Closet Door Stopper

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Section A

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Project Description

Going into this project, I knew I wanted to create something functional for my dorm room. While I didn't need anything at the time, my roommate mentioned that her right closet door could not close fully and had to have something placed in front of the door to prevent it from swinging open. Thus came the idea for a device that would hold the door in place, but also be adjustable to allow the door to open when needed.

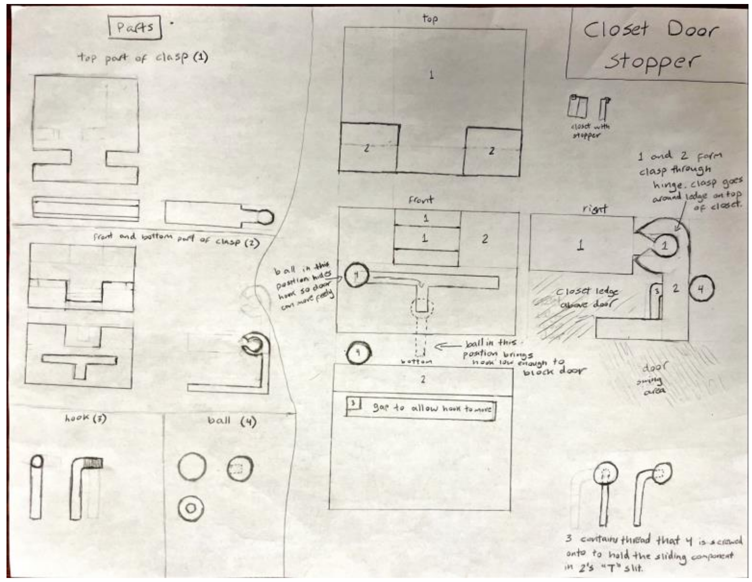


Figure 1: Ideation Sketches

My design incorporates a special sliding hook and ball feature to accomplish this feat. Although the final sliding slot changed from my original design, the principle is the same. By inserting the hook through the slot and fastening the ball to the short end of the hook, the hook can only move along the slot's track. When the hook is moved to the bottom part of the slot, the bottom of the hook will stick out and block the closet door from swinging. To free the door, the hook just needs to be moved out from the bottom of the slot.

Required CAD Features

Top of Clamp

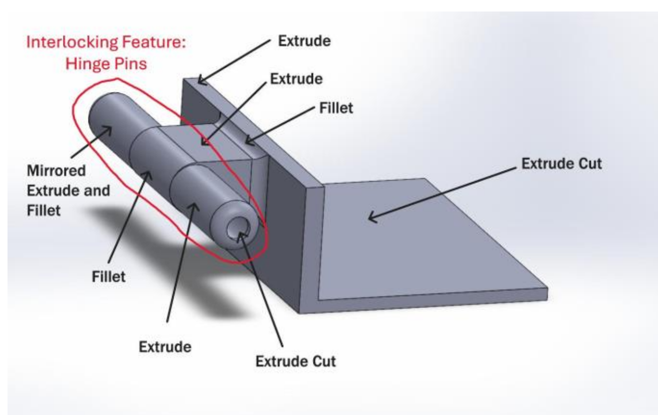


Figure 2: Top of Clamp Required Features

- Extrusion: Base
- Extruded Cut: Cut Base into "L" shape
- Extruded Cut: Bottom of Base
- Extrude: Extension of Base for Pegs
- Extrude: Right Peg
- Fillet: Right Peg
- Mirror: Right Peg
- Fillet: Front of Extension of Base for Pegs
- Shell: Pegs and Front Part of Base
- Extruded Cut: Hollow Out Pegs and Extension of Base for Pegs
- Fillets: Smooth connection between extension of base and base
- Extruded Cut: Hollow out extension of base to allow for powder removal

Front of Clamp:

- Extrusion: Front part of clamp
- Extrusion: Bottom part of clamp
- Shell: Bottom part of clamp
- Extrusion: Left C part of Hinge
- Intersect: Remove front of clamp above the C part of the hinge
- Extrusion: Right C part of Hinge
- Intersect: Remove front of clamp above the C part of the hinge
- Extrusion Cut: Slot in front of clamp for hook to slide through
- Extrusion Cut: Gap in bottom of clamp for hook to slide through
- Fillets: Smoothen slot in front of clamp
- Text: On front of clamp to provide function instruction
- Chamfers: Back of bottom of clamp
- Fillet: Round the bottom of the front of clamp

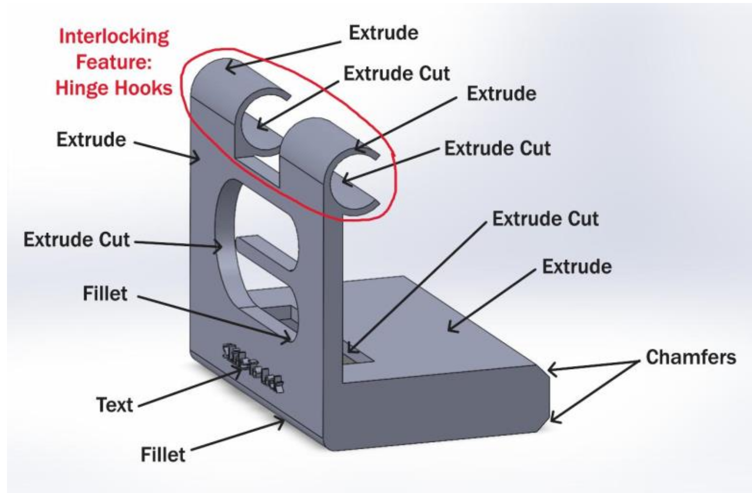


Figure 3: Front of Clamp Required Features

Hook:

- Sweep: Body of hook
- Shell: Hollow out body of hook
- Extrude: Top cantilever
- Mirror: Bottom cantilever
- Fillet: Smoothen bottom of hook

Figure 4: Hook Required Features

Ball:

- Revolve: Body of ball
- Extrusion Cut: Outside rim of ball
- Extrusion Cut: Inside of ball
- Text: Label “Door Stopper”
- Fillet: Smoothen outer rim to allow cantilevers to slide in more easily

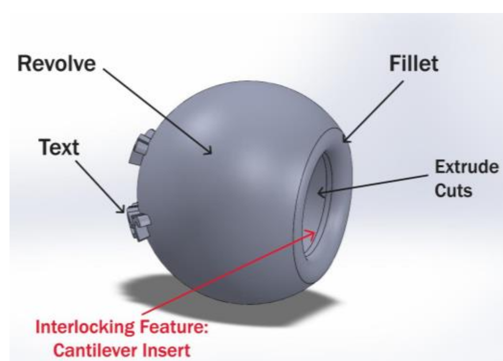


Figure 5: Ball Required Features

Interlocking Features

The front of the clamp has C hooks that attach to the pins on the top of the clamp, forming a hinge joint. This hinge allows the closet doorstopper's position to be adjusted or removed easily. The hook has cantilevers that slide into the ball. Cantilevers provided a stronger connection and made more sense for the ball-hook system as these parts would not be separated often, and the system needed to move as one piece after the initial connection.

Interlocking Features Tolerances

For the hinge joint, I chose a tolerance of 0.012 inches because the hinge is not meant to move much, and the PowerPoint presentation recommended a 0.008-0.012 inch size difference. As I wanted the diameter of the hinge joint to be 0.4 inches, I decreased the diameter of the pegs by 0.012 inches to 0.388 inches and increased the diameter of the C hooks to 0.412 inches. Similar steps were taken for the sliding slot for the hook. The recommended tolerance range for mechanically active parts was 0.016-0.020 inches, and I picked the higher end of the tolerance to ensure the parts would not be too small. To have the hook have a 0.4 inch diameter, I increased the vertical width of the sliding slot by 0.020 inches to 0.420 inches. For the cantilevers, I made sure that the ball's hole was deep enough for the hook section of the cantilevers to fit in. I also checked that the diameter of the rim was smaller than the height of the cantilevers, and that the hole inside of the ball had a diameter just slightly larger than this height. I did not reference the recommended tolerances for the cantilever snap-fit and picked numbers that I felt made sense and looked like a good fit in the assembly.

Design for Manufacturing Considerations

When designing the clamping mechanism that holds onto the top of the closet ledge, I split the clamp into two separate parts – one that goes on top and one that is in front and goes underneath the ledge – because I wanted the whole device to be easily removable. With this design in mind, I also considered which fastener would keep the front of the clamp held in place without much lateral movement with the top of the clamp while still being easily detachable and settled on a hinge joint. As for the hook and ball, I needed a fastener that would keep the ball firmly attached to the hook so that when the ball was moved, the ball and hook would move together as one. Originally, I decided to use a thread for the connection, but due to the size of the hook, a thread would be difficult to create without destroying the structural integrity of the hook. Therefore, I switched to adding cantilevers on the hook and hollowing out the ball since this attachment would be secure, and I didn't anticipate the ball-hook system being removed from the overall device once assembled. For both the hinge joint and the cantilevers, I took inspiration from the example fasteners pictured in the PowerPoint presentation.

Size of the parts was a limiting factor in my project. Because the height of the closet ledge was already smaller than 3 inches, I had no trouble designing my parts to be within the 3x3x3 cubic inches boundary. However, because my parts needed to match actual dimensions in order to fit around the closet, several of the features were constrained to a maximum size that restricted what designs were feasible. For instance, the height of the front of the clamp without the C hooks could not be too tall, otherwise the bottom part of the part would not be able to fit in the gap between the top of the closet and the closet door. This height constraint subsequently prevented the sliding slot for the hook on the front of the clamp from being larger than the height constraint minus 0.5 inches (the fixed height of the section of the front of the clamp that juts out). Since the hook's diameter needed to closely match the width of the sliding slot, the hook's diameter was limited as well; this limitation led to my switching of a thread to a cantilever fastener on the hook as I mentioned earlier.

For potential problem areas, I found that putting my parts into an assembly helped troubleshoot known issues and exposed issues that I did not know about. One issue I anticipated might arise from the printed parts was that the cantilever snap fit would be too large to fit in the hole as I was not confident about the flexibility of the 3D printed material. Using the interference detection in SolidWorks allowed me to test different-sized cantilevers to ensure that the cantilevers would fit inside of the ball's hole once inserted. The interference feature also highlighted that the gap between my C hooks on the front of the clamp was the same as the section of the top of the clamp that would be in between the hooks. This discovery led me to build in tolerance for the gap because otherwise, the C hooks would not be able to rotate around the pegs.

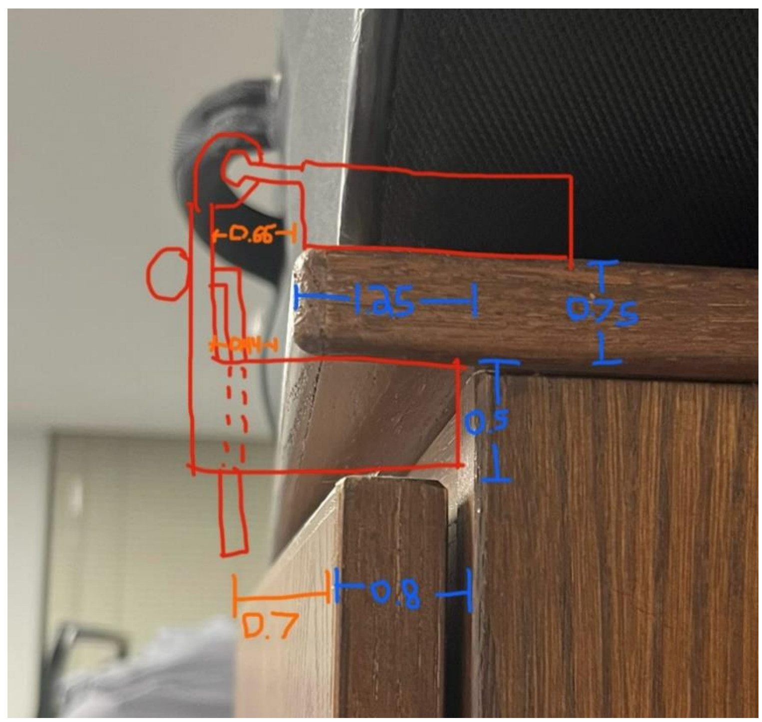


Figure 6: Closet Dimensions in Inches

After the general shapes of my parts were completed, I analyzed which parts could be hollowed and where powder might become trapped. I changed the top of the clamp, which was at first a large rectangular prism, to form instead an "L" shaped extrude since I realized my roommate stores her suitcase on top of her closet, and the suitcase could act as a counterweight to hold up the door stopper. I also hollowed out several of the other parts such as the pegs on the top of the clamp, the portion of the front of the clamp that goes underneath the closet ledge, and the inside of the hook.

My design was generally based on a clamp as the device would stay in place due to its tight fit around the top and bottom of the closet ledge. The specific part dimensions were based on measurements I took of the closet, and if there was no real dimension to match, I chose a value that made sense for the part's purpose and the overall size of the build.

Challenges Faced in Manufacturing

All my parts printed with their features as designed. I did not have difficulty assembling the pieces, but because I designed all of my tolerances to be at the maximum tolerance, some of the connections were better than others. For instance, the hook's cantilevers easily flexed inwards to lock into the ball. However, the ball would only stay fixed to the hook if the ball's hole was concentric with the hook's face. If the ball or hook was at a slight angle, the cantilevers would be able to slide out of the ball. Because the ball was prone to detaching from the hook, the hook-ball system also had difficulty staying inside the sliding slot in the front of the clamp. In my CAD model, I should have either made the cantilevers larger, the rim of the ball larger, or reduced the amount of tolerance for the cantilever joint. The hinge joint was also a looser fit than preferred, but the connection was much better than the cantilever joint. After snapping the C hooks onto the pins, I was able to move the front of the clamp to any orientation, and the hinge joint would keep the part in that orientation.

If I were to remake this design to speed up the process, I think I would combine the top of the clamp and the front of the clamp into one part. I designed them to be separate to reach the interlocking features requirement, but the closet doorstopper could be made a lot easier if the hinge joint was removed. The purpose of the hinge joint was to allow for adjustments in the distance between the top and bottom of the clamp in case the parts did not print out to fit the closet, but from seeing how accurate the 3D printer was, I think the hinge joint is not necessary.

Another takeaway from this project is that the nominal dimensions a design is based on will be different from real dimensions of the actual design. Many factors can cause this discrepancy depending on the manufacturing process. In this case, the quality of the 3D printer was the main decider. Since our projects were printed with an EOS Formiga P11, the difference between the nominal and real dimensions would be minimal because the printer was very precise. As a result, the amount of tolerance needed for connecting parts was also minimal.

Tolerance Analysis

Feature	Dimension in CAD	Actual Measured on 3D part	% difference
Top of Clamp: Peg Diameter	00.388 inches	00.405 inches	04.28752%
Top of Clamp:	00.90 inches	00.911 inches	01.1248%

Distance between Pegs			
Front of Clamp: C Hooks Outer Diameter	00.512 inches	00.500 inches	02.37154%
Front of Clamp: C Hooks Inner Diameter	00.412 inches	00.432 inches	04.73934%
Front of Clamp: Distance between C Hooks	00.91 inches	00.906 inches	00.440529%
Front of Clamp: Width of Sliding Hole for Hook	00.42 inches	00.437 inches	03.96733%
Front of Clamp: Width of Slot in Bottom of Part for Hook to Move In (from front to back)	00.44 inches	00.452 inches	02.69058%
Hook Diameter	0.4 inches	0.377 inches	05.92021%
Hook Cantilevers: Outside Distance between Beam (non-hook sections)	00.24 inches	00.252 inches	04.87805%
Hook Cantilevers: Outside Distance Between Outermost Hook Sections	00.38 inches	00.375 inches	01.3245%
Hook Cantilevers: Inside Distance between Cantilevers	00.10 inches	00.094 inches	06.18557 %
Hook Cantilevers: Height of Hook Section	00.14 inches	00.142 inches	01.41844%
Hook Cantilevers: Cantilever Length from Front of Hook to Tip of Cantilever	00.40 inches	00.4265 inches	06.41258%
Ball: Outside Diameter	00.75 inches	00.745 inches	00.668896%
Ball: Inside Diameter of Hole	00.36 inches	00.352 inches	02.24719%
Ball: Total Depth	00.42 inches	00.4375 inches	04.08163%
Ball: Rim Depth of Hole	00.069 inches	00.066 inches	04.44444%

Table 1: Final Design's Dimensions

As shown by the values in the table above, the 3D printer was very accurate in producing the CAD models nearly exactly to their dimensions. All my tolerances for the interlocking features had a percent difference of less than 6.5% from the desired dimension. This difference was only a couple thousandths of an inch at most and mostly occurred in the hook. The length of the cantilevers, measured from the front of the hook to the end of the cantilevers, had the greatest disparity of 6.41% with an actual length of 0.4265 inches when the length was supposed to be

0.4 inches. The inside distance between the cantilevers was the second-greatest difference of 6.18% with an actual distance of 0.094 inches as opposed to the CAD dimension of 0.1 inches. The smallest difference came from the distance between the C hooks on the front of the clamps. With just a 0.44% difference, the actual printed part was 0.906 inches, extremely close to the intended 0.9 inches.

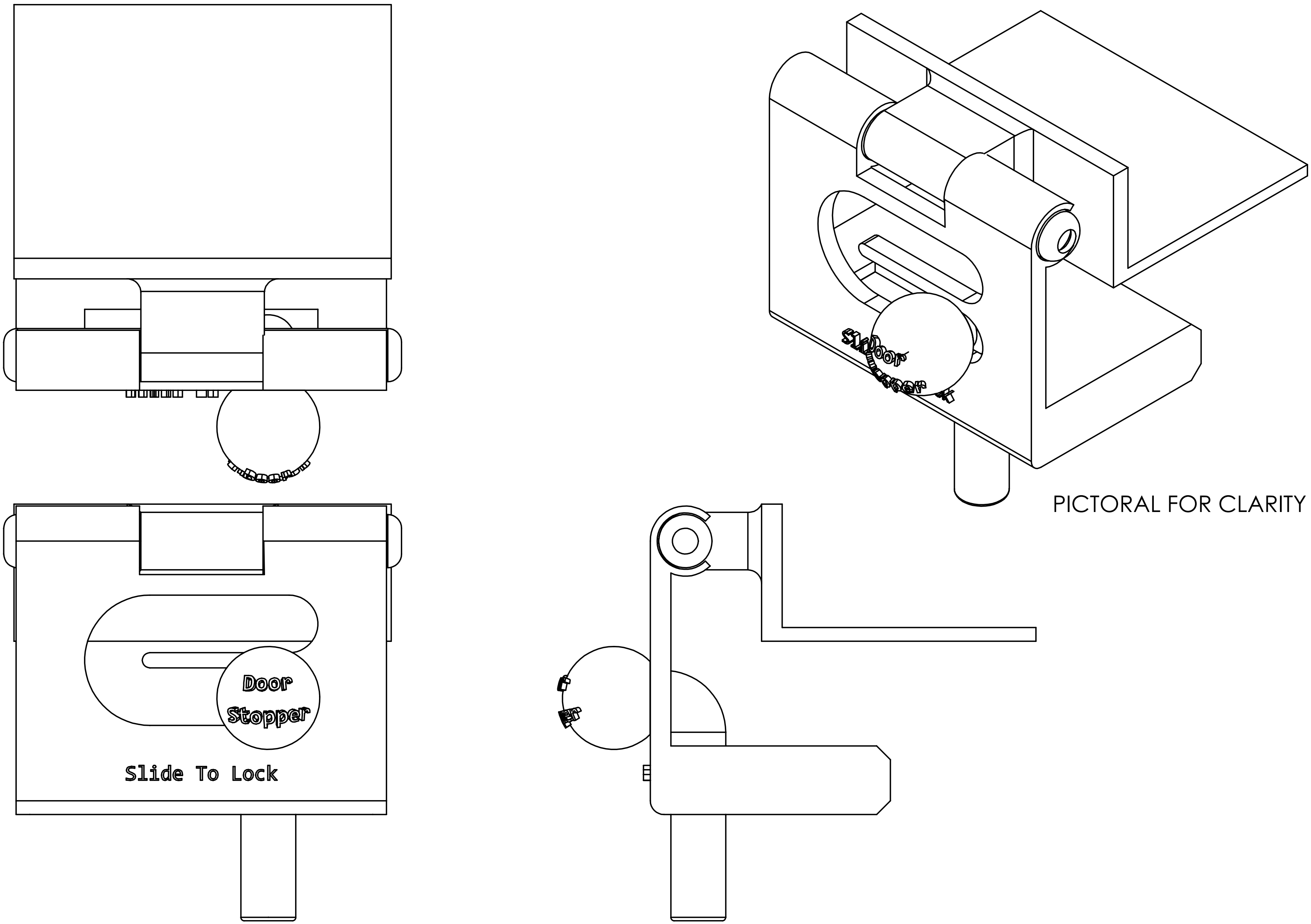
Analyzing the data, there does not appear to be a correlation between when a part will print to be larger or smaller than its digital component as 9 of my measured tolerances were larger, and 8 were smaller. While assembling my pieces, I noticed that my parts all connected without any issue; however, the fits tended to be on the looser end. From tightest to loosest: the hook inside of the sliding slot in the front of the clamp, then the hinge joint between the pins and the C hooks, and lastly, the cantilever joint between the hook and the ball. Although not a perfect fit, this tolerance was not big enough for the pieces to come loose by themselves. Thus, my closet doorstopper was able to function as designed.

To obtain better interlocking features in the future, especially if my parts were going to be printed with as accurate of a machine as an SLS printer, I would choose a tolerance closer to the medium range of tolerances as opposed to the higher range of tolerances.

Conclusions

In order, I created the top of the clamp, the front of the clamp, the hook, and the ball. Although creating each part was quick because I had thought a lot beforehand about what each part should look like, identifying design issues was easier after I could see the parts in their entirety. Once these issues were resolved, I made alterations to reduce the amount of powder that might get trapped in my design. To do this step, I looked for parts that were solid but could serve the same function if hollow, and if the part needed to be solid, I added a hole for powder to be removed. The final touches were to remove sharp edges that someone handling the parts might come in contact with. From this project, I learned that no matter how much planning you put into a design, creating a perfect product on the first try is almost impossible. The closet doorstopper assembled together without any issues, fit on the closet as I wanted it, moved as intended, and yet, the 3D printed product was too light to handle the door's weight properly. The engineering design process is iterative for a reason, and this project has given me valuable experience with the first stages of designing a product that I can now employ in the future.

CAD Drawings



Georgia Institute of Technology	TITLE **ASSEMBLY**		DATE **11/26/2024**	NAME **JOY D'SOUZA**			
	COURSE **ME 1670**	SEMESTER **FALL 2024**	SECTION **A**	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES SCALE 1:1 SHEET 1 OF 1	SIZE **A**

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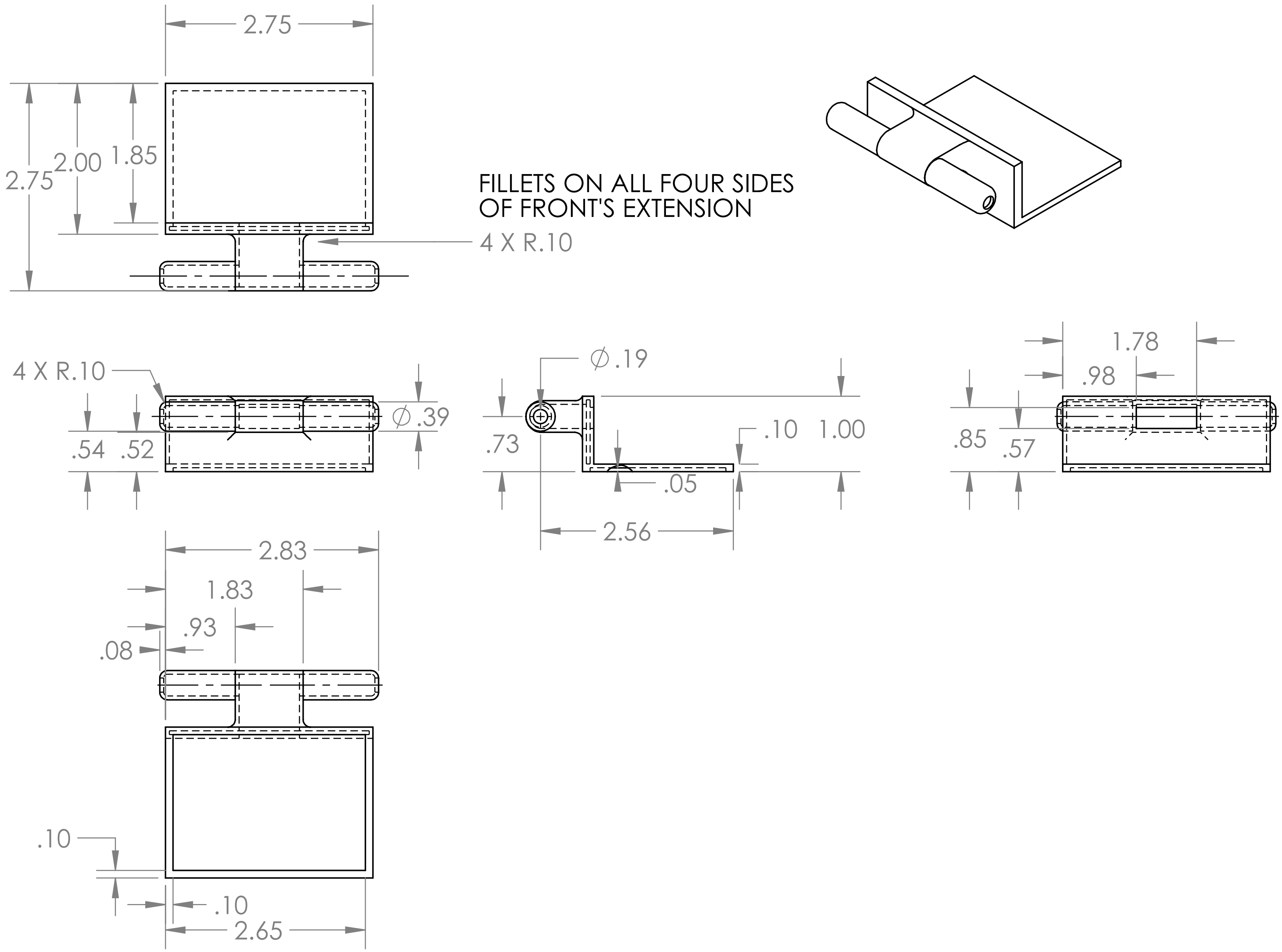

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COURSE ME 1670	SEMESTER FALL 2024	SECTION A	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES	SIZE A
					SCALE 1:2 SHEET 1 OF 1	

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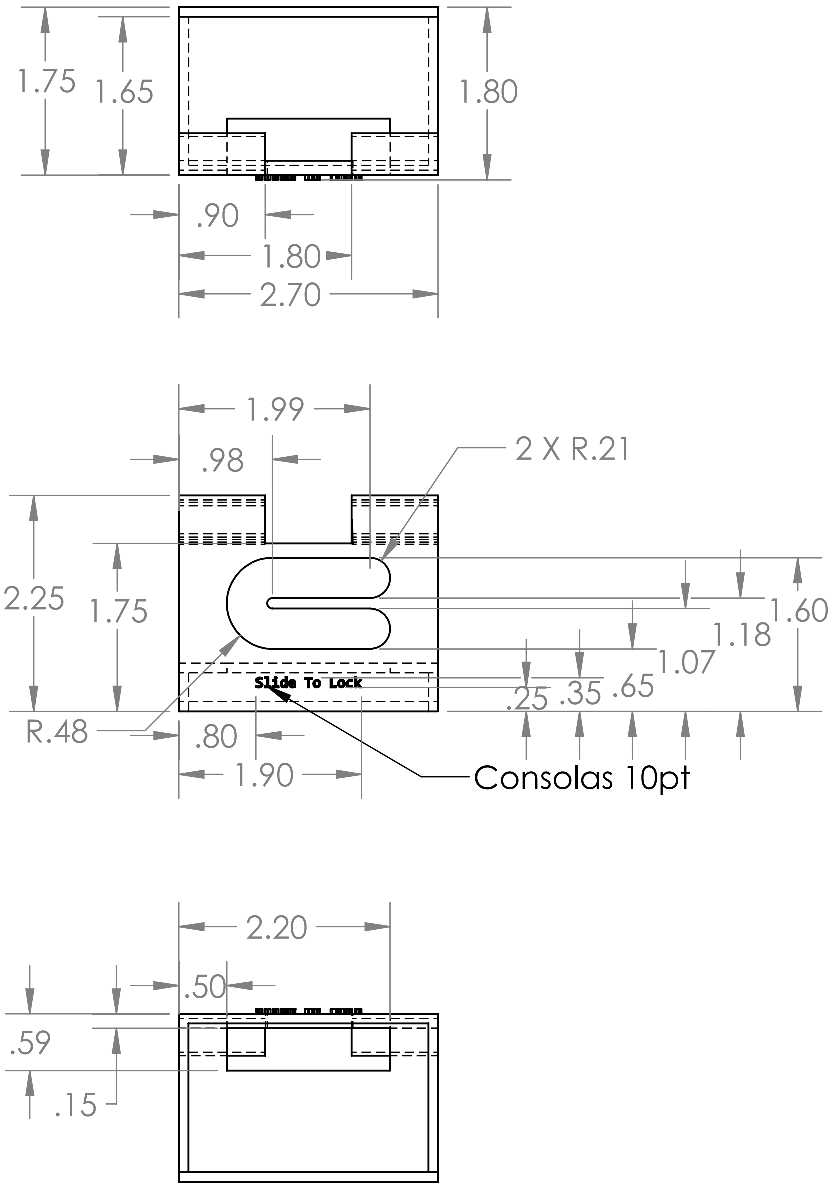

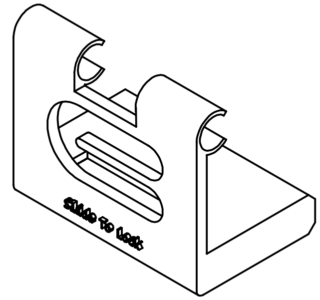

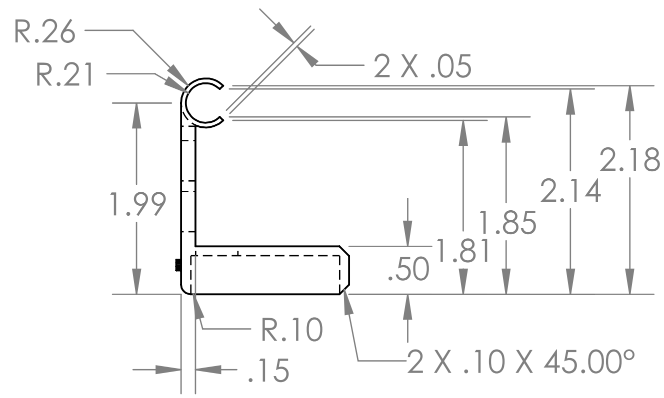

UNIFORM WALL THICKNESS OF 0.1 INCHES,
UNLESS OTHERWISE SPECIFIED

Georgia Institute of Technology

TITLE		DATE	NAME
FRONT OF CLAMP		11/26/2024	JOY D'SOUZA

COURSE	SEMESTER	SECTION	LAB	ACTIVITY		SIZE
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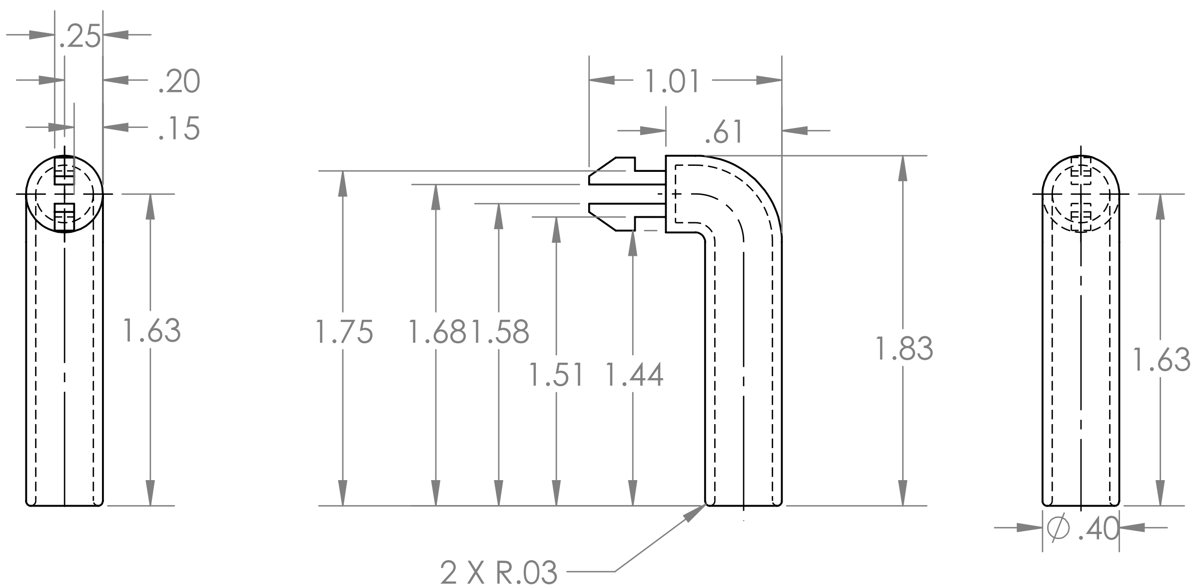

UNIFORM WALL THICKNESS OF 0.05 INCHES,
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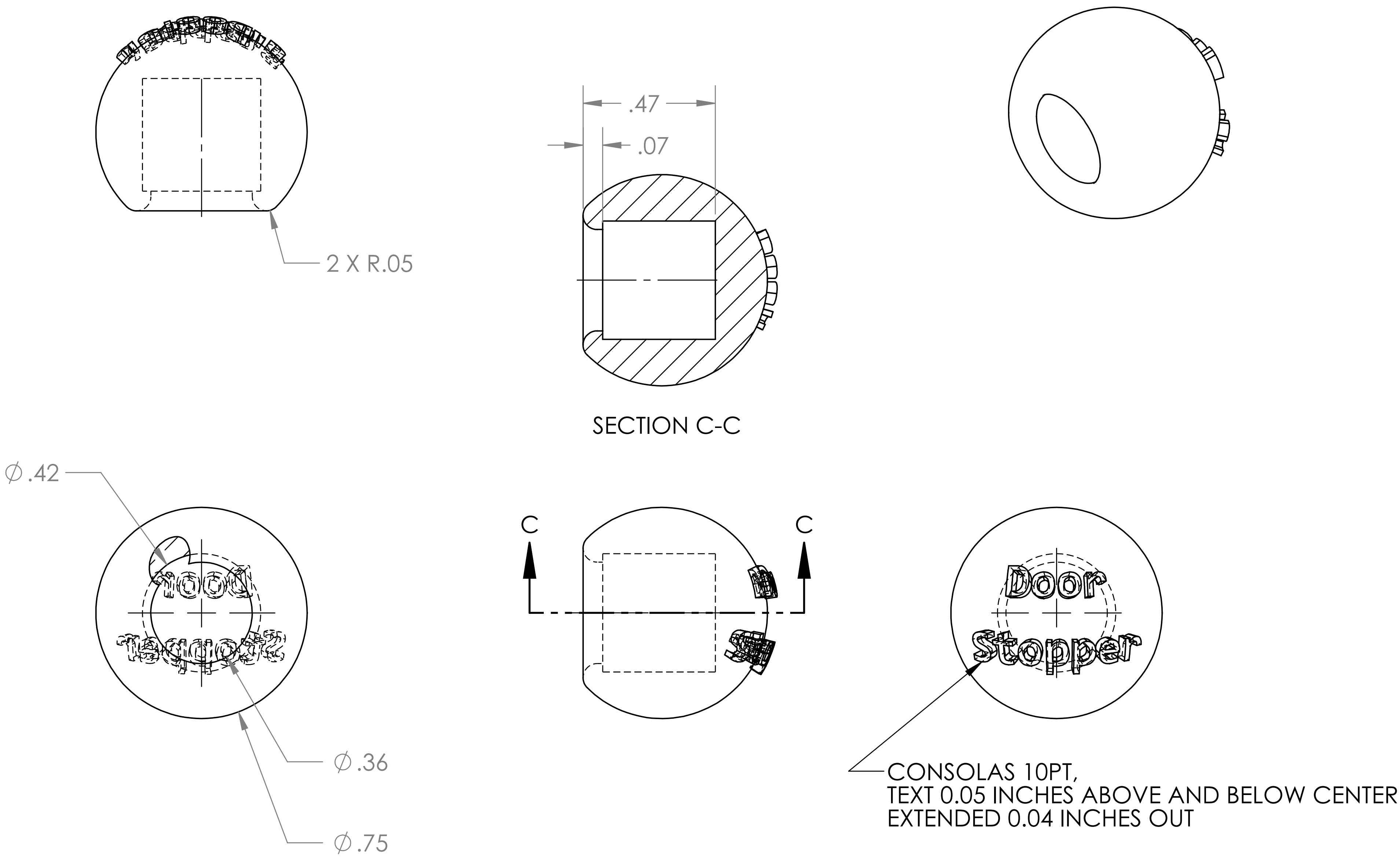
Georgia Institute of Technology	TITLE: HOOK		DATE: 11/26/2024	NAME: JOY D'SOUZA		
	COURSE: ME 1670	SEMESTER: FALL 2024	SECTION: A	LAB	ACTIVITY	DIMENSIONS ARE IN INCHES ANGLES ARE IN DEGREES; SCALE 1:1; SHEET 1 OF 1; SIZE: A

5 4 3 2 1



Georgia Institute of Technology

TITLE		DATE	NAME		
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5 4 3 2 1